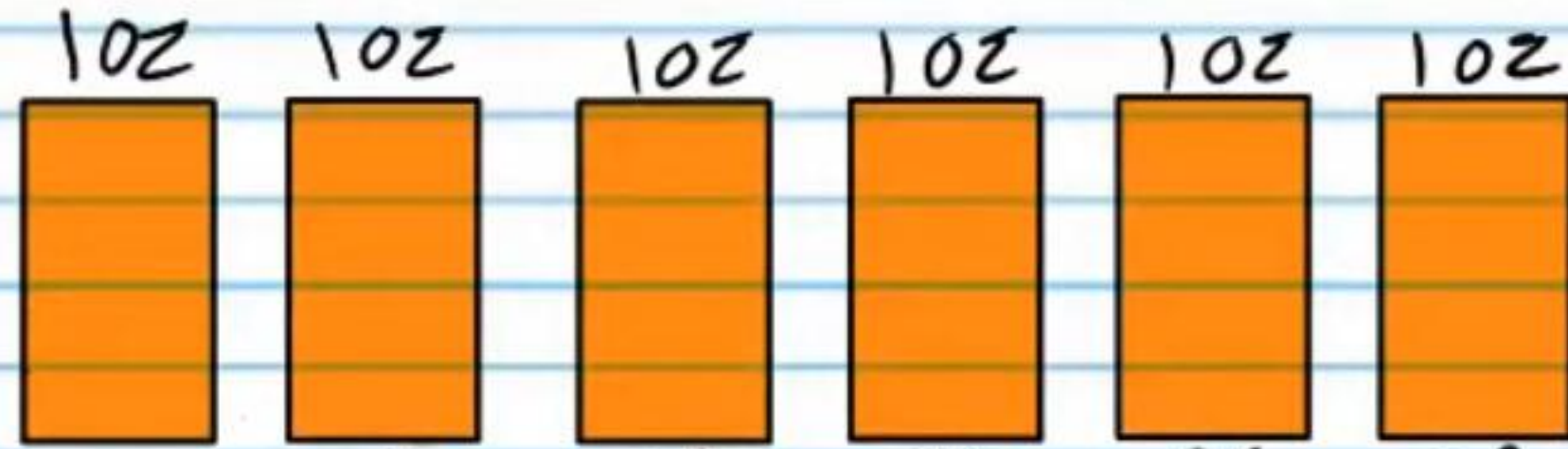


## Class 22: Applications of Percents

Example: Find 23% of 6.



$$\frac{23}{100} + \frac{23}{100} + \frac{23}{100} + \frac{23}{100} + \frac{23}{100} + \frac{23}{100} = \frac{138}{100} = 1 \frac{38}{100}$$

① What is 7% of 0.91? ② Find 5% of 19.

③ 6.7% of 8 is what number? ④ What is 3% of 28?

⑤ Find 92% of 56. ⑥ What is 18% of 7.5?

⑦ 0.004% of 0.06 is what number? ⑧ What is 79% of 0.0035?

⑨ 143% of 42 is what number? ⑩ What is 6.6% of 111?



### Type II Percent Problems: Finding the Original Number

⑪ 75 is 5% of what number? ⑫ 89% of what number is 40.05?

⑬ 192 is 6% of what number? ⑭ 4% of what number is 0.316?

⑮ 8.28 is 23% of what number? ⑯ 0.55% of what number is 78?

### Type III Percent Problems: The Percent as the Unknown

⑰ 36 is what percent of 45? ⑱ What percent of 9 is 52?

⑲ 24 is what percent of 75? ⑳ What percent of  $\overline{8}$  is 3?

㉑ 5.87 is what percent of 6? ㉒ What percent of 0.2 is 0.08?



②③ 0.006% of the human body is iron. If a man has 0.0075 pounds of iron in his body, how much does the man weigh in total?

②④ If 65% of the human body is oxygen, how much of a 79 pound boy is oxygen?

②⑤ If a 9 pound sample contains 2.3481 pounds of sulfur, what percent of the sample is sulfur?

②⑥ If 8 grams of lithium hydride contains one gram of hydrogen, what percent of the lithium hydride is hydrogen?

②⑦ Bill was given 60% of the tuna catch. If Bill was given 80 pounds of tuna, how much tuna was caught in total?

②⑧ If 9 karat gold is 37.5% pure gold, how much of a 0.004 pound 9 karat gold necklace is pure gold?



②⑨ If 88 percent of apple juice is water and there are 6 ounces of water in a cup of apple juice, how many ounces of apple juice are in the cup?

③⑩ If 65% of a human body is oxygen, and there are 110.5 pounds of oxygen in a man, what is the total weight of the man?

③⑪ There is an 8% sales tax on a T-shirt. If the T-shirt costs \$98, how much money will a purchaser have to pay for tax?

③⑫ If a customer pays \$1.12 for an 8% sales tax on a T-shirt, what was the price that the store listed for the T-shirt (i.e. the price before tax)?

③⑬ 3.2% of a human is nitrogen. If a woman weighs 99 lbs, how much of her is nitrogen?

③⑭ If 3% of an animal is nitrogen, and an animal contains 1.86 pounds of nitrogen, what is the total weight of the animal?



③⑤ If 88 percent of apple juice is water, how many ounces of water are contained in 9 ounces of apple juice?

③⑥ If a person drinks six grams of tea, and there are 0.03 grams of sugar dissolved in the tea, what percent of the tea is sugar?